



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester (CSE/IT/AIML/IOT/MC/MV/ADS/PI)

Mid Term – I Examination

Date: February 20, 2025

Max. Marks: 15

Time: 8:00 am – 8:50 am

Paper: Integral Calculus and Difference Equations (MAIC 102)

Instructions:

- Answer all the questions.
- The calculator is not allowed into the examination hall.
- Start writing your answers from the first page. Do not leave it blank. No extra answer sheets will be provided.

Q1.

Find a power series solution of $8x^2 \frac{d^2 y}{dx^2} + 10x \frac{dy}{dx} - (1+x)y = 0$ about $x = 0$. (5M)

Q2.

Find all the eigenvalues and eigenfunctions of the Sturm-Liouville boundary value problem $y'' + \lambda y = 0$, $y(-\pi) = y(\pi)$, $y'(-\pi) = y'(\pi)$. (5M)

Q3.

Evaluate the double integral $\int_{y=0}^1 \int_{x=y}^1 e^x dx dy$. (5M)